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Xinyu Guo Ph.D. student

Education

University of Arizona Arizona, United States
Ph.D. Student in Computer Science 2024 - Present
- Advisor: Prof. Mihai Surdeanu
- Research Interests: LLM Reasoning, RL Credit Assignment, Reinforcement Learning, Interpretability

Shandong University Shandong, China
B.E. in Computer Science 2019 - 2023
- GPA: 89.56/100.00
- Graduated with Honors

Research Overview

My research focuses on **LLM-based reasoning**, along two directions:

- **RL Credit Assignment for Reasoning Supervision**: outcome rewards tell a model *whether* it reasoned well, but not *where*. I assign credit at a finer granularity: to the key phrases in a reasoning trace, using rewards derived from the model's own correct predictions (CogRE), and to the tokens responsible for each rubric criterion, using rubric-conditioned critics (DVE).
- **Reasoning Mechanisms**: looking inside LLMs to understand how concepts and their logical structure support reasoning, by aligning sparse autoencoder features with concepts (AlignSAE) and taxonomies (TaxoSAE).

Publications

RL Credit Assignment for Reasoning Supervision

[2026] **Xinyu Guo**, Zhengliang Shi, Minglai Yang, Mahdi Rahimi, Mihai Surdeanu*. The Answer Lies Within: Self-Derived Rewards Enable Explainable Relation Extraction. ***EMNLP 2026 Main.***
University of Arizona, Shandong University.

[2026] **Xinyu Guo**, Zhengliang Shi, Minglai Yang, Chen Xu, Lingyong Yan. Disentangled Value Estimation: Rethinking Scalar Critics in Rubric-Based RL. *Under Review.*

Reasoning Mechanisms

[2026] **Xinyu Guo**, Minglai Yang, Zhengliang Shi, Clayton T. Morrison, Eduardo Blanco, Mihai Surdeanu. TaxoSAE: Rethinking Sparse Autoencoders through Taxonomic Logic. *Under Review.*

[2026] Minglai Yang, **Xinyu Guo**, Zhengliang Shi, Jinhe Bi, Steven Bethard, Mihai Surdeanu, Liangming Pan. AlignSAE: Concept-Aligned Sparse Autoencoders. ***TMLR 2026.***
University of Arizona, Shandong University, Peking University, LMU Munich.

Other Publications

[2026] Minglai Yang, **Xinyu Guo**, Utkarsh Tyagi, Mian Zhang, Razvan Dumitru, Sunjie Hou, Yunzhong He, Daniel Yue Zhang, Ying Liu. Rubric Dropout: A Simple Way to Mitigate Reward Hacking in Rubric-as-Reward RL. *arXiv 2026.*
Scale AI, University of Arizona, University of Texas at Dallas.

[2026] Minglai Yang, **Xinyu Guo**, Renxiong Wang, Daniel Yue Zhang, Yunzhong He. From Copying to Composition: How Language Models Develop In-Context Learning. *Under Review.*

[2026] Minglai Yang, Xinyan Velocity Yu, Pengyuan Li, **Xinyu Guo**, Zhenting Qi, ..., Zexue He. Dr. DocBench: A Comprehensive Benchmark for Expert-Level and Difficult Document Parsing. *NeurIPS 2026 Workshop LCFM*.

[2026] Yunze Xiao, Jienan Chen, **Xinyu Guo**, Minghao Liu, Zihan Wang, Weihao Xuan. RecuritOpsBench: When Agents Fail to Ask for the Information They Need. *Under Review*.

[2025] Zhengliang Shi, ..., **Xinyu Guo**, ..., Zhaochun Ren. Deep Research: A Systematic Survey. *arXiv 2025*.
Shandong University, University of Arizona, CMU, Tsinghua University, etc.

[2025] Zhaoze Sun, Chengliang Chai*, Qiyan Deng, Kaisen Jin, **Xinyu Guo**, Han Han, Ye Yuan, Guoren Wang, Lei Cao. QUEST: Query Optimization in Unstructured Document Analysis. Accepted by *VLDB* 2025.
University of Arizona, MIT, Beijing Institute of Technology.

Internships

Abaka AI | Mountain View, CA, United States 2026.05 - 2026.10

- Role: Research Scientist Intern
- Built RL environments and post-training data for tool-using customer-service agents on τ^2-bench, and trained call agents with RL (RetailServe).
- Contributed to Dr. DocBench and helped organize the DocInsights 2026 workshop at EMNLP 2026.

Research Experience

CLU-LAB, University of Arizona | Arizona, United States 2025.03 - Present

- Advisor: Prof. Mihai Surdeanu
- Research Topic: Natural Language Processing, Reinforcement Learning, LLMs.

CAES Research Center, Shandong University | Shandong, China 2023.09 - 2024.09

- Advisor: Prof. Zhongxing Yu
- Research Topic: Program Analysis, Automated Program Repair, AI4SE.

Oxford Winter Exchange Program, Oxford University | UK 2022.02

- Advisor: Prof. Jeremy Pitt
- Topic: AI Myths, Reality, and Human-AI Interaction.

Grant and Services

Co-PI for ABC SEED Grant, Arizona Astrobiology Center

Reviewer for ARR 2026

Awards and Honors

Galileo Circle Scholarship, University of Arizona 2026

Science Graduate College Fellowship, University of Arizona 2024, 2025

Outstanding Undergraduate Thesis Award, Shandong University 2023

Outstanding Graduate Student, Shandong University 2023

Academic Scholarship, Shandong University 2022

The 18th Certified Software Professional Examination Top 4.57%, CCF 2021

Skills

Programming Languages: `Python`, `C/C++`, `MATLAB`, `Go`, `Java`, `HTML`.

Libraries & Framework: `PyTorch`, `Numpy`, `vLLM`, `VeRL(RL)`, `Hugging Face Transformers`, `SQLite`, `MySQL`, `Spring Boot`, `JavaScript`, `Qt`.

Teaching

Teaching Assistant for CSC337 Web Programming, CSC460 Database
University of Arizona, US 2024, 2025

Teaching Assistant for Software Engineering
Shandong University, China 2023